

VALLABH NITURKAR

Backend Engineer | Distributed Systems & Financial Technology

Pune, India | +91-7558591138 | vallabhwork18@gmail.com

linkedin.com/in/vallabhniturkar | github.com/vpndev18

SUMMARY

Backend engineer with 3 years at **FIS Global** building production .NET services for a distributed task-execution platform that schedules and runs **1.2M+** financial workloads per day across Kubernetes engine nodes. Depth in concurrency, message-driven reliability and autoscaling for regulated banking systems, with full-stack and AI delivery on top.

TECHNICAL SKILLS

Languages: C#, C++, SQL, TypeScript, JavaScript

Backend: .NET 8/9, ASP.NET Core, Minimal APIs, EF Core, MediatR, SignalR, REST APIs, xUnit

Concurrency & Messaging: Multithreading, Lock-Free Primitives (Interlocked), RabbitMQ, Redis, Event-Driven Architecture, Microservices

Cloud & DevOps: Docker, Kubernetes, Azure, CI/CD (GitHub Actions), Git

Databases: SQL Server, PostgreSQL

Frontend & AI: React, TypeScript, RAG Pipelines, Qdrant (Vector DB), Gemini AI

PROFESSIONAL EXPERIENCE

Software Engineer | FIS Global, Pune, India

Sept 2023 – Present

*Distributed task-execution platform for **payments** workloads — schedules, routes and executes large-scale asynchronous jobs across **40+** engine nodes. Tech: C# / .NET 8, ASP.NET Core, RabbitMQ, Redis, PostgreSQL, Docker, Kubernetes (Helm, KEDA).*

- Cut broker round-trips on transient task failures by **~35%** by designing a bounded in-process retry queue with backpressure and metrics, replacing full **RabbitMQ** re-queue cycles; validated under **64-thread** stress tests at **1,500 tasks/sec**.
- Eliminated lock contention on the task-execution hot path (**150 tasks/sec** per engine at peak) with a lock-free usage tracker built on **Interlocked** atomics — now the data source behind engine recycling and health monitoring.
- Owned the engine health endpoint backing **Docker HEALTHCHECK** and **Kubernetes** liveness/readiness probes, the prerequisite that unblocked the platform's **KEDA** autoscaling rollout across **40+** nodes.
- Removed **~12 manual engine restarts per week** by shipping a threshold-based recycling policy (task count + runtime) that retires long-running engines without dropping in-flight work, closing a reliability gap open across **4 releases**.
- Migrated **3 payments** modules from **.NET Framework** to **.NET 8**, added **Minimal APIs** for new internal endpoints, and cut high-volume **T-SQL** procedure runtime from **~45s to ~6s** through targeted indexing.
- Run the platform on production **on-call rotation** — triaged **30+** Sev-2/Sev-3 incidents and review **~10 PRs/week** across the engine and scheduling services.

PROJECTS

Foresight – AI Financial Forecasting Platform | github.com/vpndev18/Foresight-AI

.NET 9, Minimal APIs, MediatR, EF Core, React, PostgreSQL, Gemini AI, Docker, GitHub Actions, xUnit, JWT

- Built a full-stack forecasting platform on **Vertical Slice Architecture** with **MediatR**, running 1,000-iteration **Monte Carlo simulations** that return P10/P50/P90 outcome bands in **~400 ms**.
- Built the **React** front end — simulation setup, JWT session flow, and result views rendering the P10/P50/P90 bands returned by the forecast API.
- Exposed the backend via **Minimal APIs (.NET 9)** with **PostgreSQL/EF Core** migrations and **Gemini AI** risk summaries; containerized with **Docker Compose** and automated **CI/CD** in **GitHub Actions** (xUnit, lint/build, secret scanning).

FinQuery – AI-Powered Personal Finance Tracker | github.com/vpndev18/FinQuery

.NET 8, React, Qdrant Vector DB, Redis, SignalR, Docker, GitHub Actions, Serilog, JWT

- Implemented a **RAG pipeline** over **Qdrant** vector search and embeddings, enabling natural-language queries across **10k+** indexed personal-finance transactions.
- Layered **Redis** caching on hot-path queries and **SignalR** server-push for live budget alerts, served through a **React** natural-language query interface.

EDUCATION

Vishwakarma Institute of Technology, Pune – B.Tech in Information Technology | CGPA: 8.85/10

Aug 2019 –

May 2023

ACHIEVEMENTS

- **International Publication:** Improved Helmet Detection Model Using YOLOv5 — published by **Springer** after international peer review (doi.org/10.1007/978-981-99-0071-8_4).
- **Programming:** 400+ problems solved across LeetCode and GeeksforGeeks.
- **Academic:** Top 1% rank in **MHT-CET 2019** among **390,000+** candidates.